AWS NETWORKING PROJECT

VPC PEERING

Private Connectivity Between Two Amazon VPCs

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Batch	Batch 1
Training Provider	EWB Cloud Labs
Region	To be confirmed for this submission
Submission Date	24 October 2026

PROJECT OBJECTIVE

Design and document private IPv4 connectivity between two Amazon VPCs using VPC peering. The workbook architecture includes two VPCs, private application workloads, explicit route tables, security groups and a bastion host for administration.

Evidence note: The screenshots supplied show AWS console forms for creating a VPC and creating a peering connection. They do not show a completed VPC creation, an Active peering status, configured routes, running EC2 instances, or successful terminal tests. These are therefore presented as configuration-stage screenshots, not proof of completed deployment.

1. PROJECT OVERVIEW

Amazon Virtual Private Cloud (VPC) provides an isolated network in AWS. VPC peering allows two VPCs to route traffic between private IP addresses when the connection is accepted and appropriate routes and security rules are configured.

Project objectives

- Use non-overlapping IPv4 CIDR ranges.
- Create public and private subnets with explicit route-table associations.
- Establish a VPC peering connection.
- Configure routes in both directions.
- Restrict SSH and ICMP traffic to the required lab hosts.
- Record actual connectivity tests and cleanup evidence.

2. RESOURCE AND IP PLAN

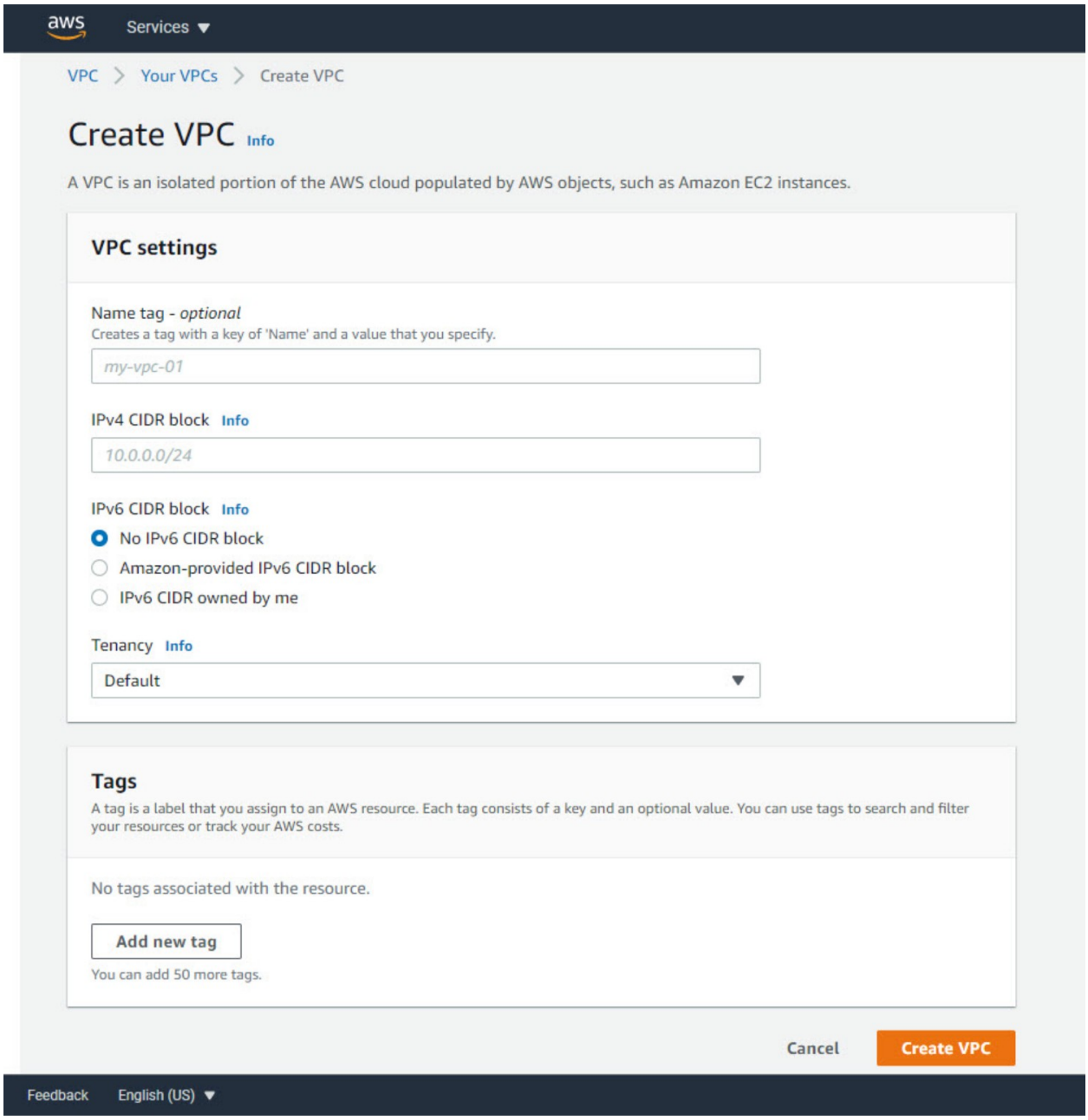
Resource	Network / subnet	IPv4 CIDR or address
ewb-vpc-a	VPC A	10.10.0.0/16
ewb-a-public	VPC A public subnet	10.10.1.0/24
ewb-a-private	VPC A private subnet	10.10.2.0/24
ewb-vpc-b	VPC B	10.20.0.0/16
ewb-b-private	VPC B private subnet	10.20.1.0/24
ewb-bastion	Public subnet A	10.10.1.10 + public IPv4
ewb-app-a	Private subnet A	10.10.2.10; no public IPv4
ewb-app-b	Private subnet B	10.20.1.10; no public IPv4

These are the planned values from the EWB workbook. Confirm that the actual AWS resources use these CIDRs before documenting the project as deployed.

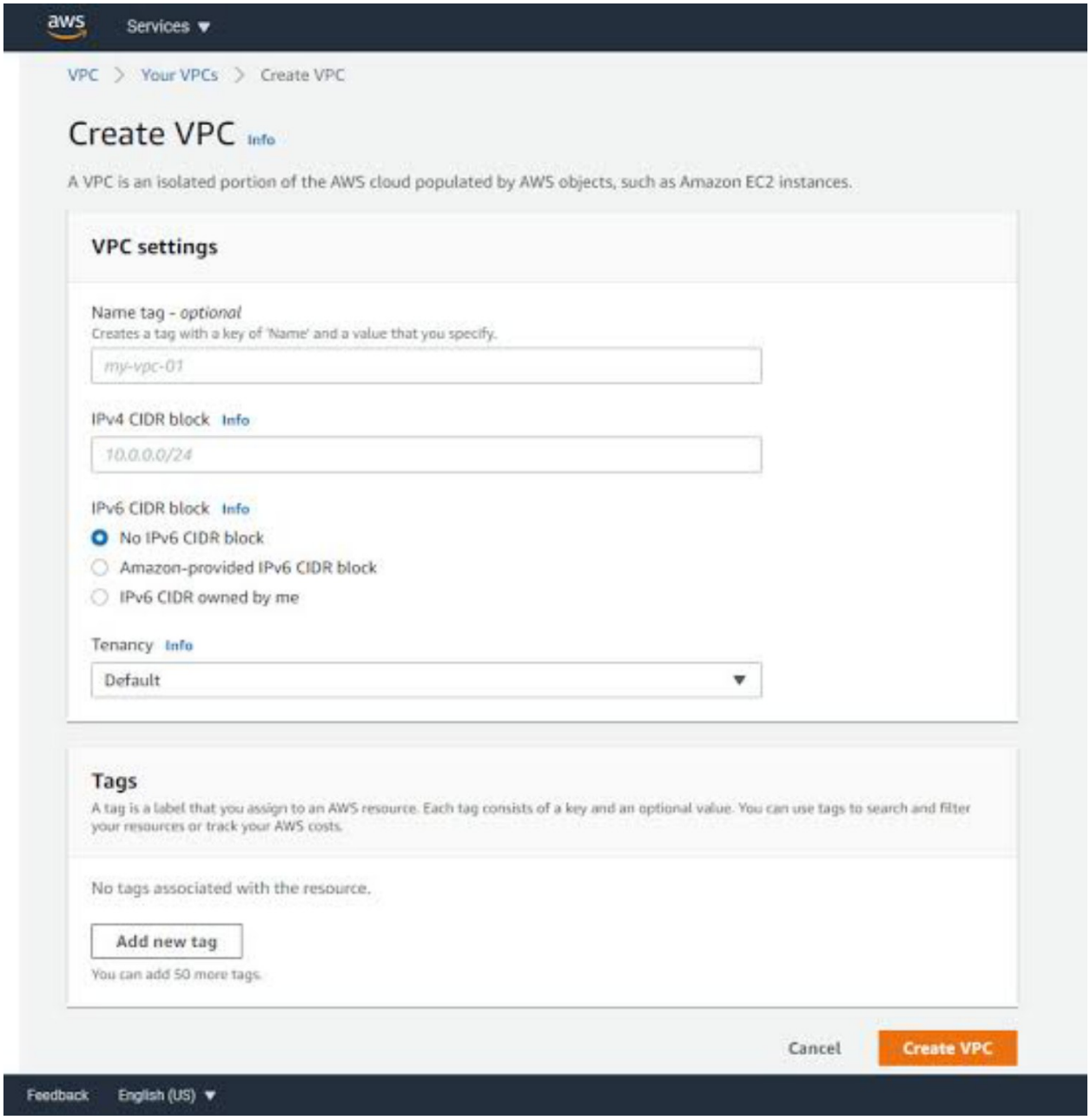
3. AWS CONSOLE SCREENSHOTS

The following evidence was supplied by the student and inserted without changing the visible AWS console content. Captions describe what each image shows.

Screenshot 1 — Create VPC form

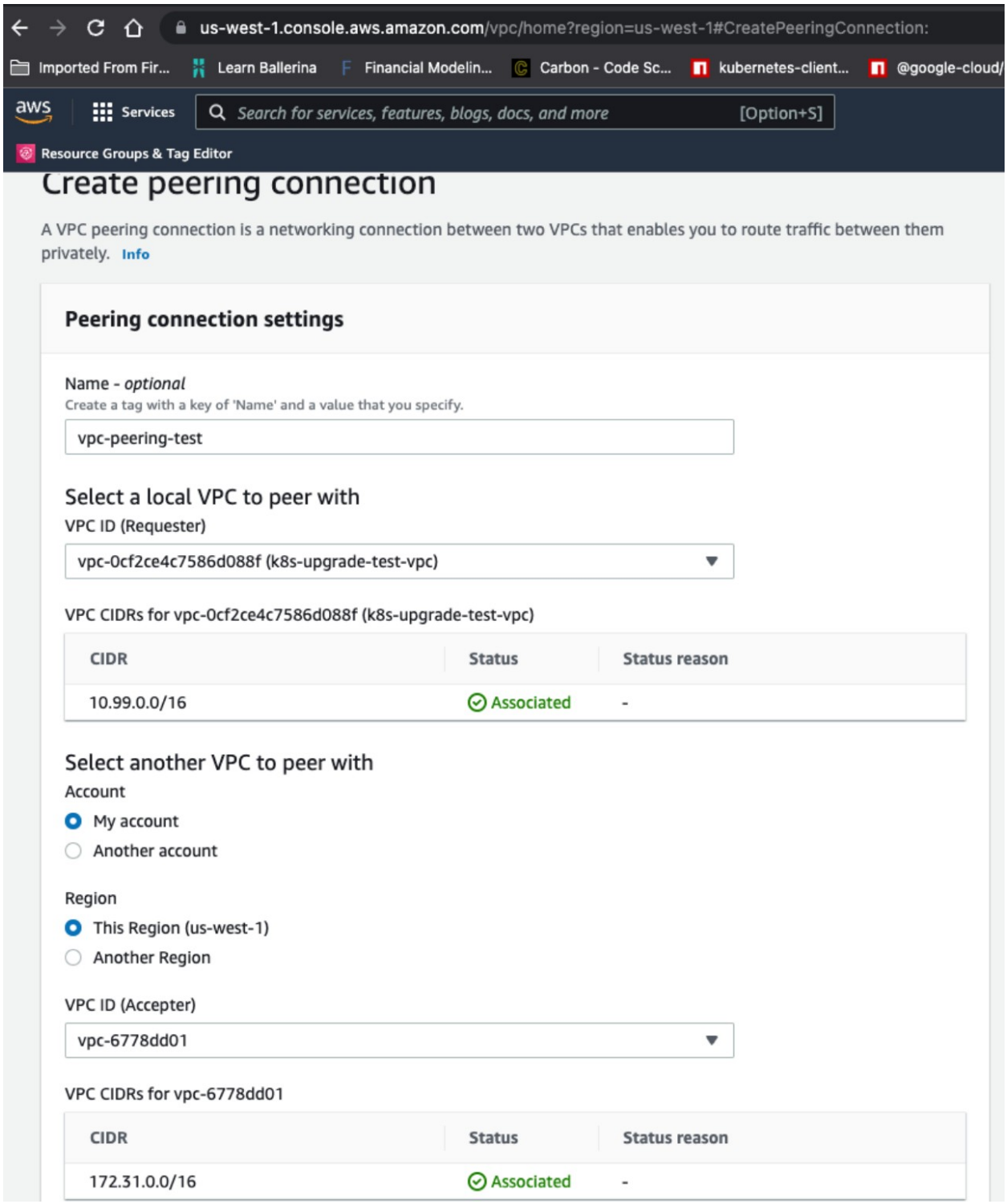


Shows the AWS Create VPC page with fields for Name tag, IPv4 CIDR, IPv6 selection and tenancy. The values shown are blank/default placeholders; this is not a completed VPC record.



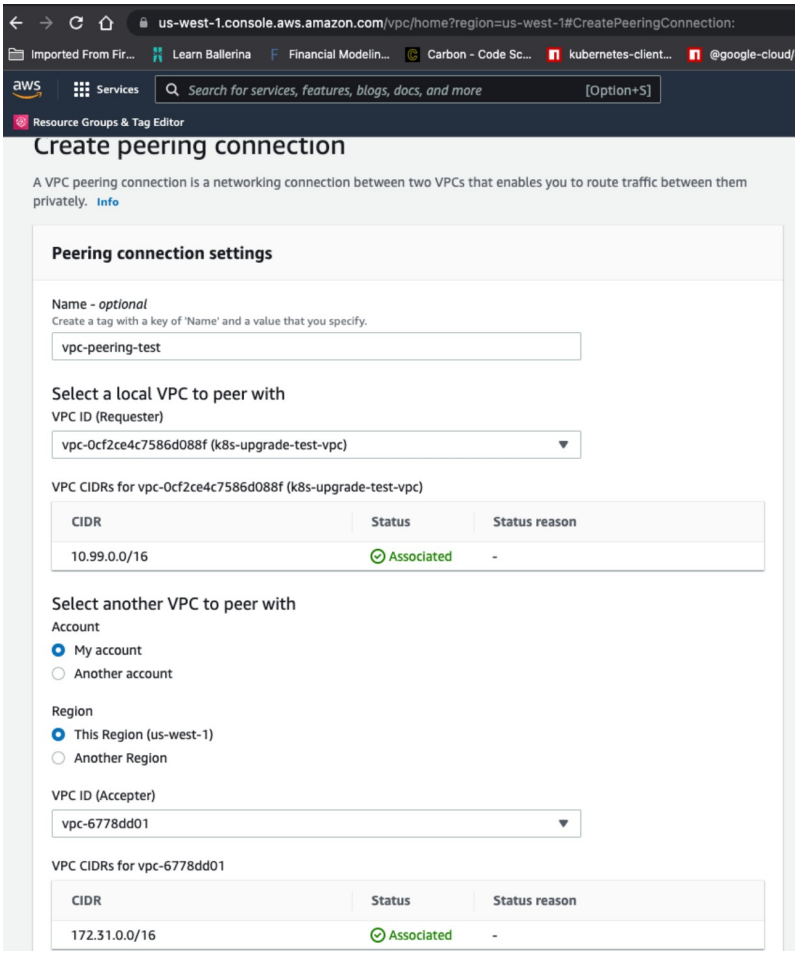
Shows the AWS Create VPC page with fields for Name tag, IPv4 CIDR, IPv6 selection and tenancy. The values shown are blank/default placeholders; this is not a completed VPC record.

Screenshot 2 — Create peering connection form



This screenshot shows a peering request form in region us-west-1 with example VPCs and CIDRs 10.99.0.0/16 and 172.31.0.0/16. These values do not match the EWB workbook plan (ap-south-1; 10.10.0.0/16 and 10.20.0.0/16). Use the workbook's values for the final project unless EWB has instructed otherwise.

Screenshot 3 — Duplicate capture



This appears to be a duplicate of the peering form screenshot above; retained as provided. Consider removing it from the final submission if only unique evidence is requested.

4. ROUTING AND SECURITY DESIGN

Route table	Destination	Target / purpose
ewb-rt-a-public	10.10.0.0/16	local
ewb-rt-a-public	0.0.0.0/0	Internet gateway ewb-igw-a
ewb-rt-a-public	10.20.0.0/16	VPC peering connection
ewb-rt-a-private	10.10.0.0/16	local
ewb-rt-a-private	10.20.0.0/16	VPC peering connection
ewb-rt-b-private	10.20.0.0/16	local
ewb-rt-b-private	10.10.0.0/16	VPC peering connection

Planned inbound security group rules

Security group	Protocol / port	Source
ewb-sg-bastion	SSH TCP 22	Your public IPv4 /32
ewb-sg-app-a	SSH TCP 22	10.10.1.10/32
ewb-sg-app-a	ICMP Echo request	10.20.1.10/32
ewb-sg-app-b	SSH TCP 22	10.10.1.10/32
ewb-sg-app-b	ICMP Echo request	10.10.2.10/32

These are proposed settings from the workbook, not verified settings from the screenshots provided. Never expose SSH to 0.0.0.0/0 and never upload private keys or AWS credentials to GitHub.

5. TEST PLAN — RESULTS TO BE FILLED

Test	Expected result from workbook	Actual result
Baseline ping before peering routes	No replies	Not provided
App-A to App-B ping after routes	4 replies; 0% packet loss	Not provided
App-B to App-A ping after routes	4 replies; 0% packet loss	Not provided
SSH to App-B via bastion	Shell on App-B	Not provided
Controlled ICMP failure and recovery	Ping stops, then resumes after rule restore	Not provided

6. CONCLUSION

The provided screenshots document the AWS console setup stage for the VPC and peering forms. The complete EWB project also requires actual resource configuration, routes, security groups and connectivity tests. Since the supplied screenshots do not establish that these later steps were completed, the deployment status and test outcomes remain unverified in this report.

7. NEXT EVIDENCE REQUIRED FOR A COMPLETE SUBMISSION

- VPC list showing ewb-vpc-a and ewb-vpc-b with the correct CIDRs.
- Subnet list and subnet-to-route-table associations.
- Peering connection showing Active status and both correct VPCs.
- All three route tables showing local and peering routes.
- Security group inbound rules.
- EC2 instance details showing App-A and App-B have no public IPv4.
- Terminal output for baseline ping, forward/reverse ping, SSH through bastion, and controlled failure/recovery.
- Cleanup evidence after removing resources created for the lab.

8. CLEANUP CHECKLIST

After capturing evidence, terminate lab EC2 instances, verify/delete associated lab EBS volumes and snapshots, release any Elastic IP, remove peering routes and the peering connection, delete lab security groups after network interfaces disappear, then remove lab subnets, custom route tables, internet gateway and VPCs. Recheck Billing after cleanup because usage reporting can be delayed.

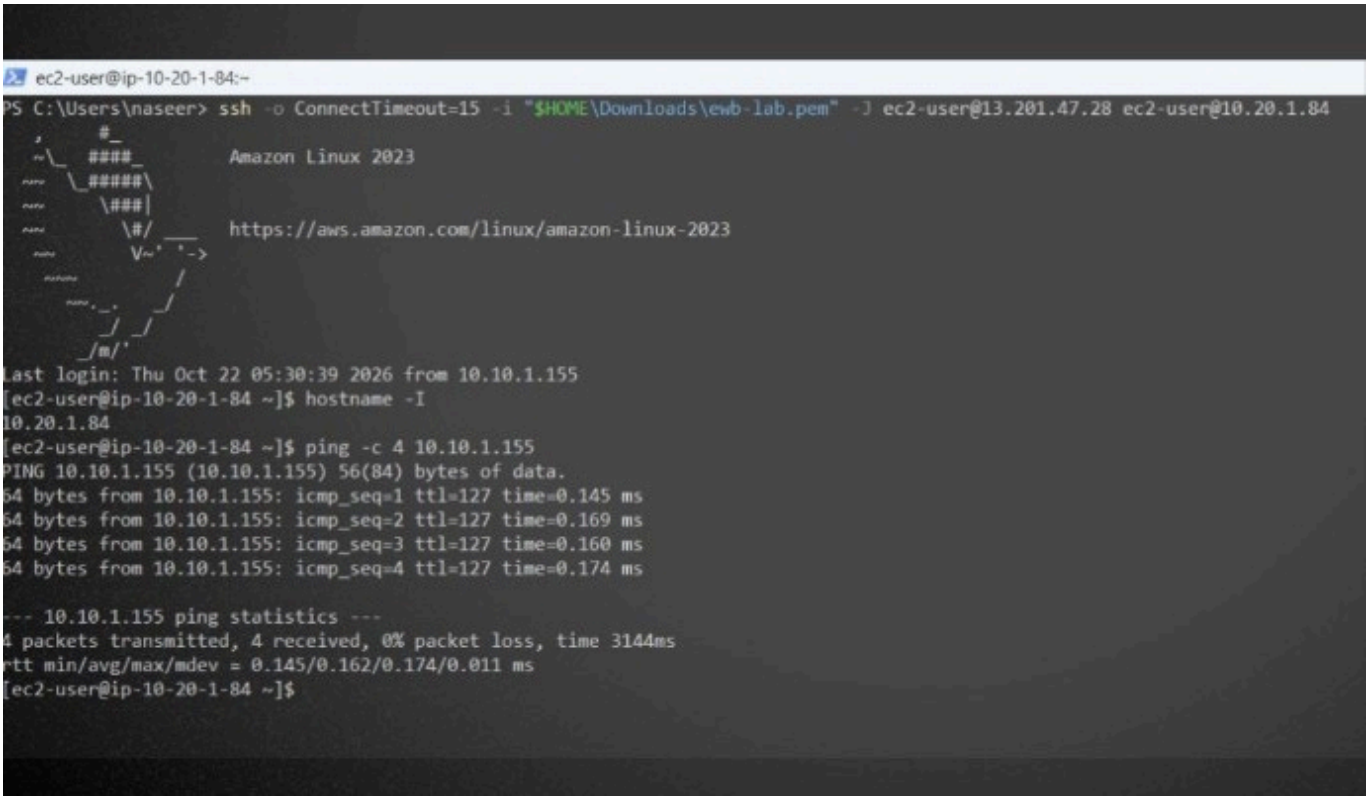
9. SOURCE

Primary source: EWB Cloud Labs, *AWS Architect Associate — Student Project Workbook: VPC Peering*, October 2026, especially pages 3–14. Planned CIDRs, route tables, security rules and acceptance criteria are based on that workbook. Console screenshots were provided by the student.

Before submission: Confirm the AWS Region and replace “not provided” only with actual results. This report does not claim successful deployment or testing that has not been evidenced.

Evidence 1 · Connectivity test from App-B

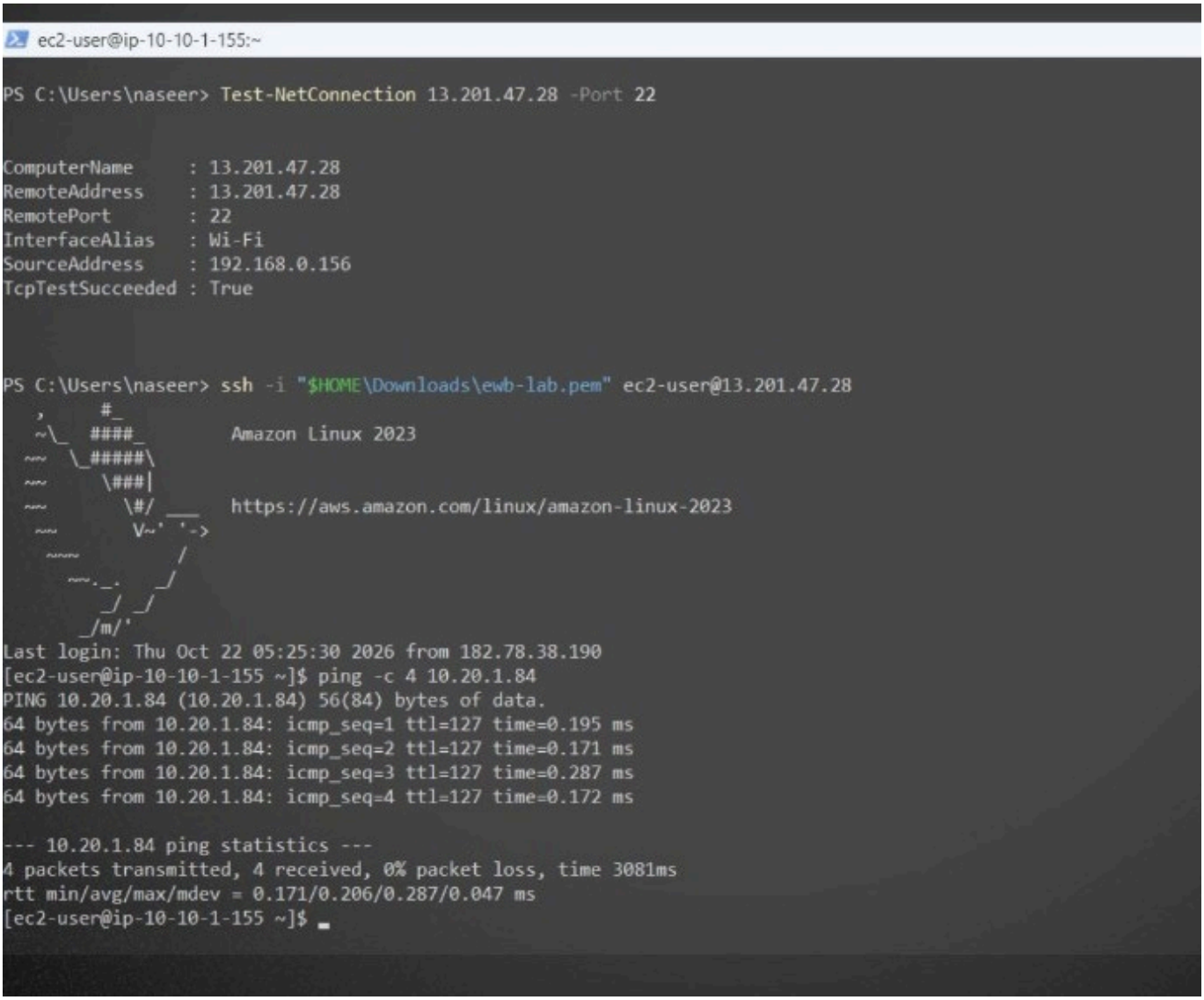
Evidence captured for report · 24 October 2026



The terminal output shows four ping replies from 10.10.1.155 with 0% packet loss.

Evidence 2 · Connectivity test from App-A

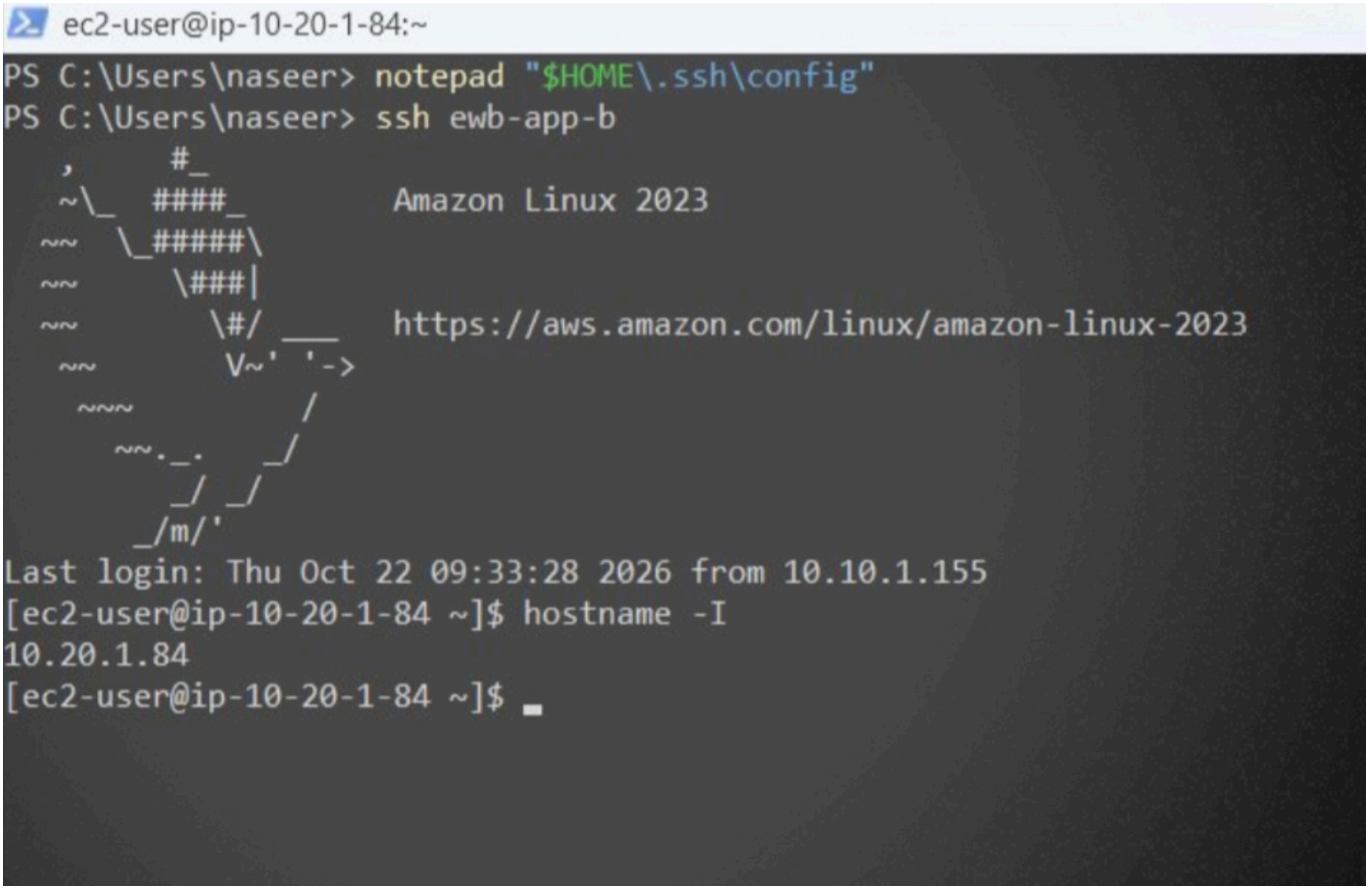
Evidence captured for report · 24 October 2026



The screenshot shows TCP port 22 connectivity to the bastion and four ping replies from 10.20.1.84 with 0% packet loss.

Evidence 3 · VPC peering architecture

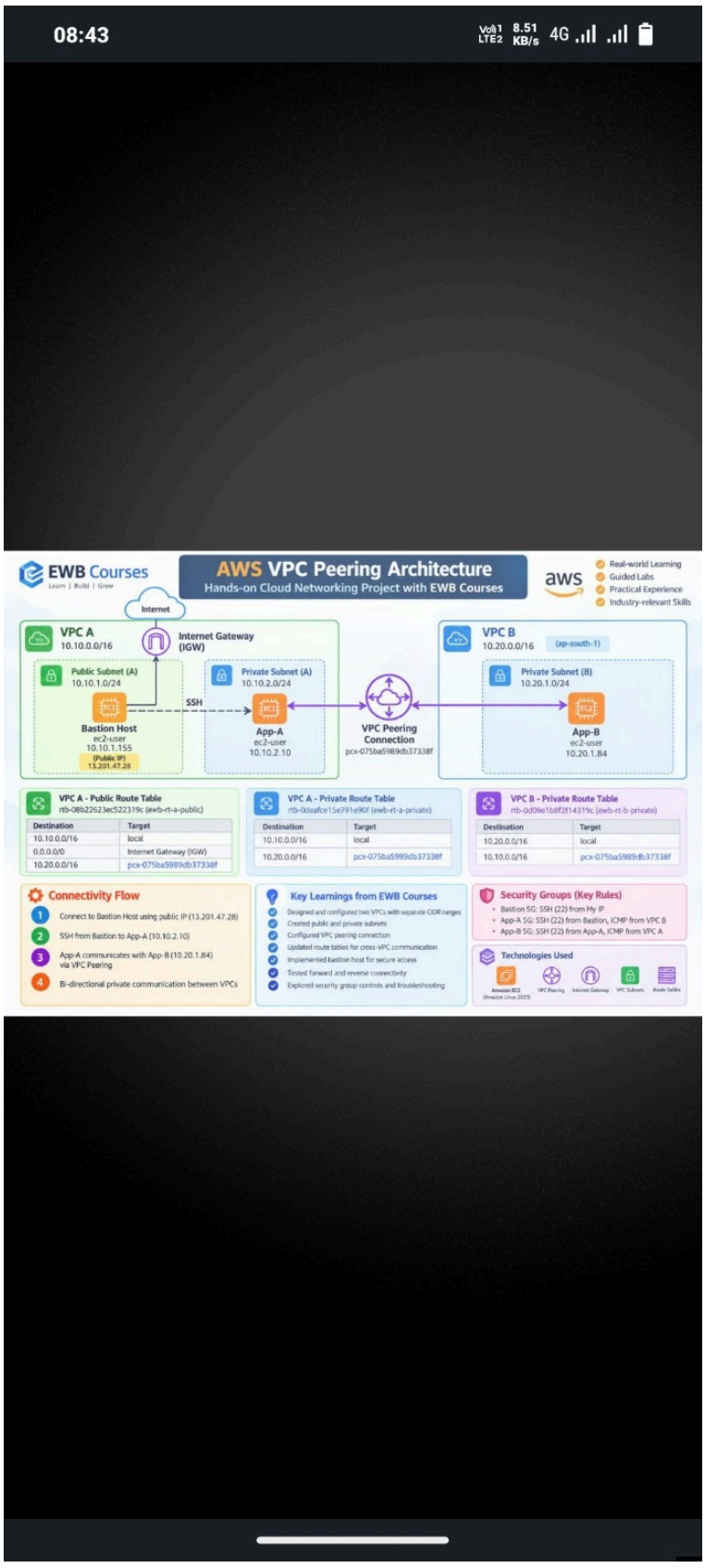
Evidence captured for report · 24 October 2026



Architecture diagram showing VPC A, VPC B, the peering connection, route tables, and security-group rules.

Evidence 4 · App-B SSH and private IP

Evidence captured for report · 24 October 2026



The terminal shows an SSH session to ewb-app-b and hostname -I output of 10.20.1.84.